Life Sciences and
Facility Management

Institute of
Computational Life Sciences

Introduction to Digital Image Processing

Part 2: Image Representation

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Motivation: Why work with digital images?

Task: What are the advantages of digital images compared to physical forms of image data (photographs, paintings, prints, murals, etc.)?

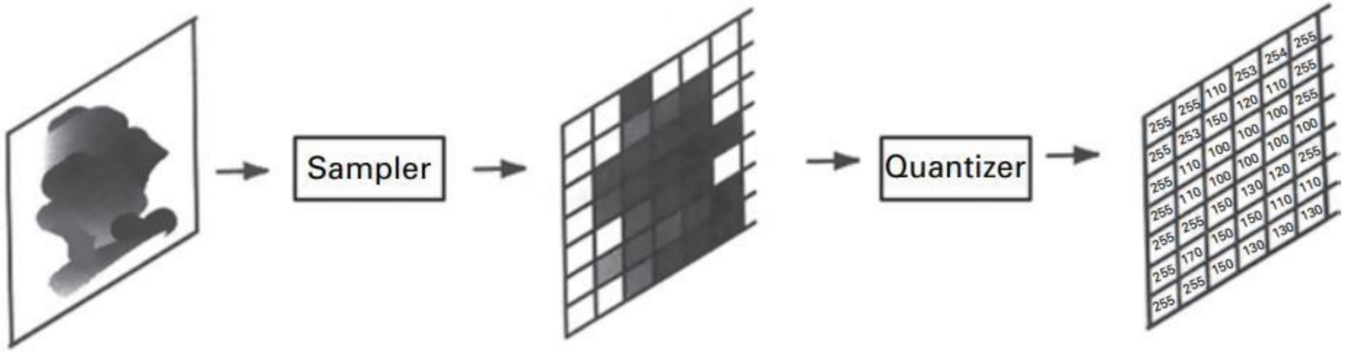
- Data storage takes no physical space
- Data can be copied and shared
- Images can be processed and analyzed
- Data visualization can be adjusted to different needs
- Digital images are much cheaper than physical copies of images

Image representation

Pixels and quantization

Objects and images

- An imaging system captures an **input signal**, such as electromagnetic radiation reflected or transmitted by an object, and converts it into an **output signal** or **image**.

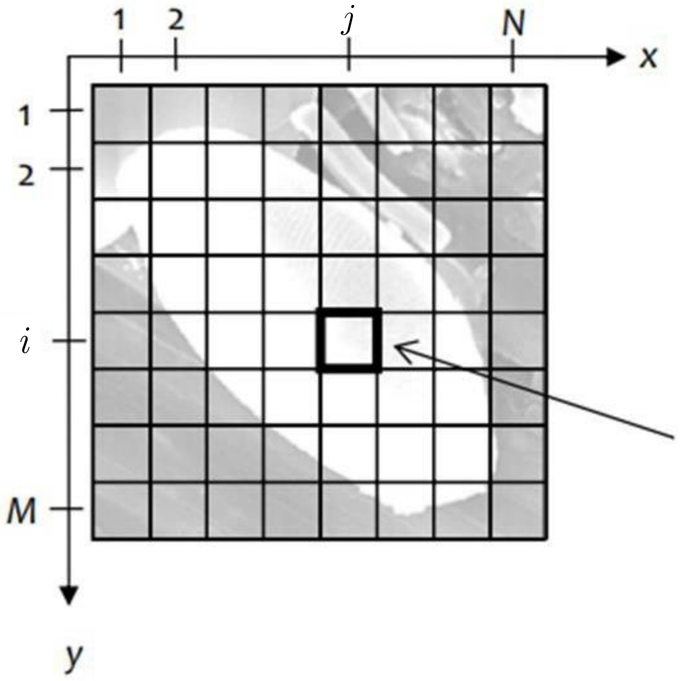


The relationship between an analog image and a digitized image.

- The advent of computers has opened up enormous new possibilities for the quantitative processing and analysis of images, provided they can be represented by **digital arrays of discrete values**.

Image data: Matrix representation

- Due to the rasterization, the digital images can be represented naturally by a matrix $f(x, y)$ or by an array.
- A picture element (i, j) is called **pixel** and the value $f(p)$ at the pixel $p = (i, j)$ is called **pixel value**



Column index $1 \leq i \leq M$
Row index $1 \leq j \leq N$

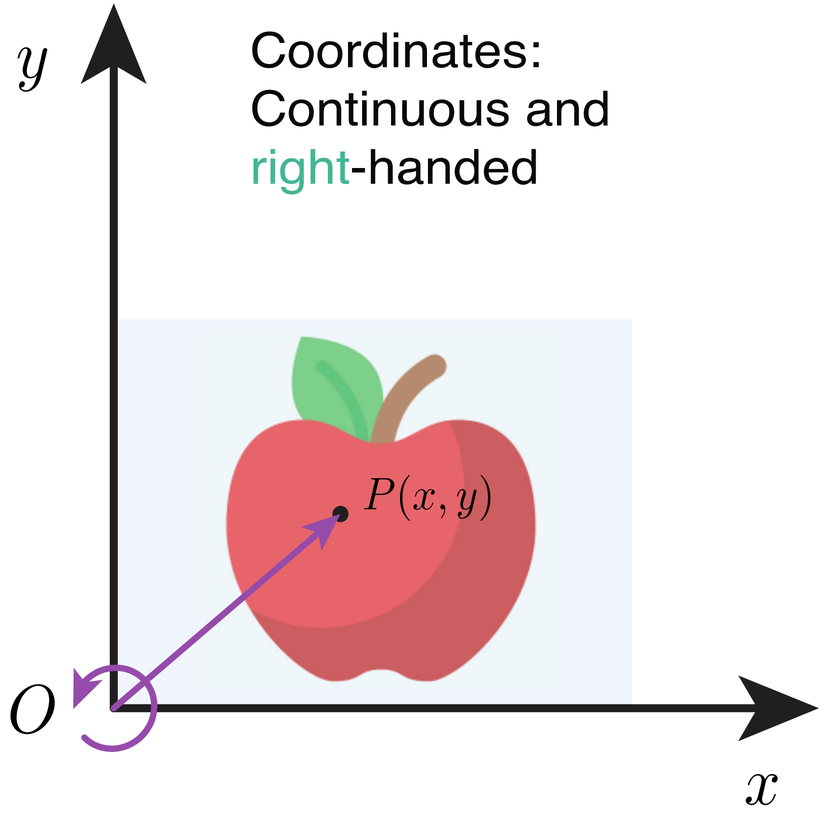
Pixel (i, j)
Pixel value f_{ij}

Matrix representation

$$\mathbf{B} = (f_{ij}) = \begin{pmatrix} f_{11} & f_{12} & \cdots & f_{1N} \\ f_{21} & f_{22} & \cdots & f_{2N} \\ \vdots & \vdots & & \\ f_{M1} & f_{M2} & \cdots & f_{MN} \end{pmatrix}$$

Coordinate systems

- 2D images are organized in 2D grids, having a horizontal and vertical extent
- To locate information in that grid requires two coordinates
- Unfortunately, we need to distinguish between different coordinate systems



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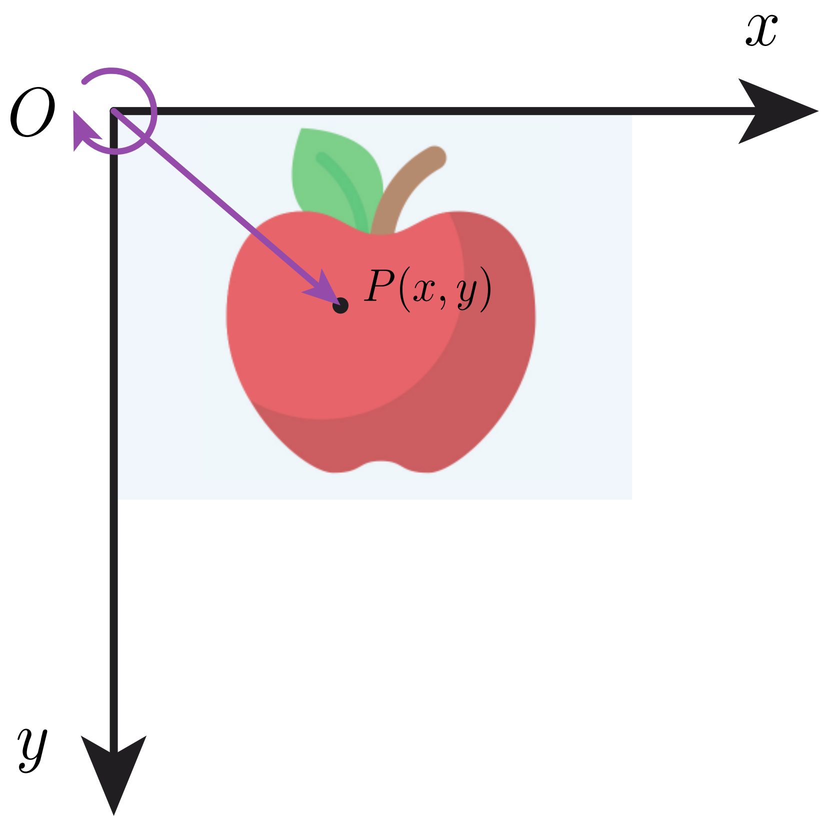
Coordinates:
Continuous and
left-handed



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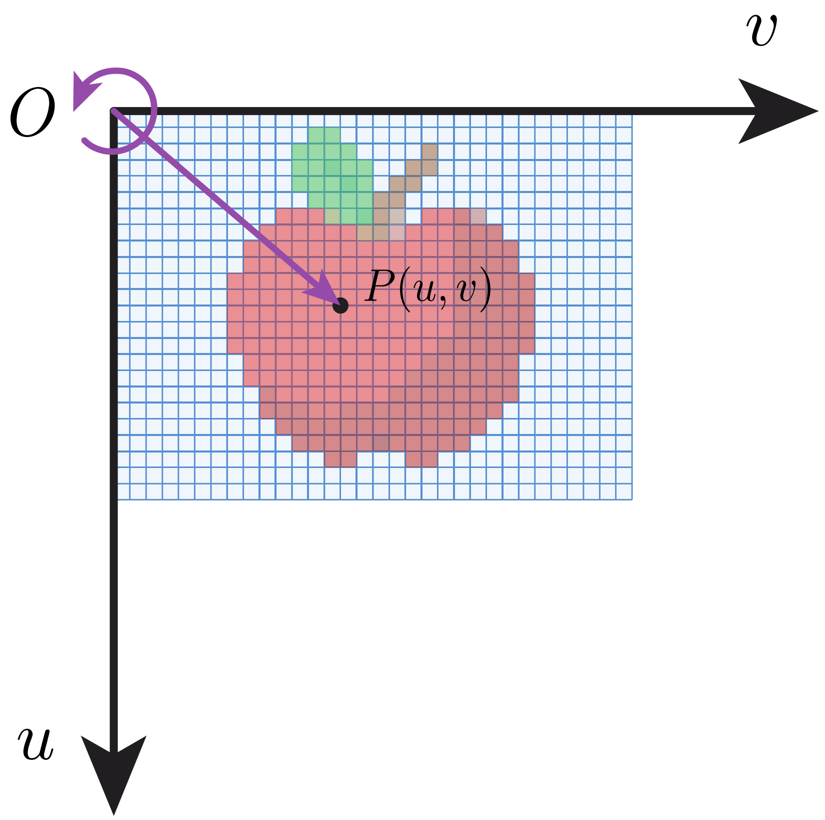
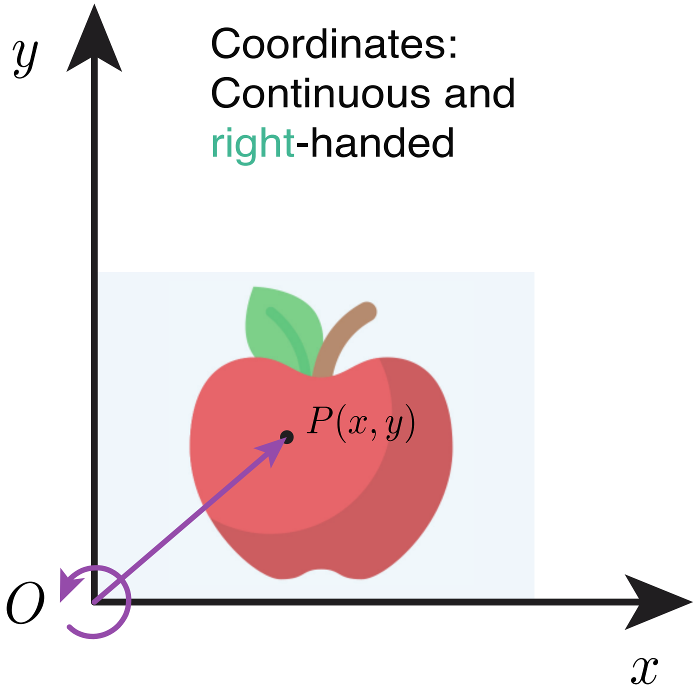
Coordinates:
Discrete and
right-handed



Coordinate systems



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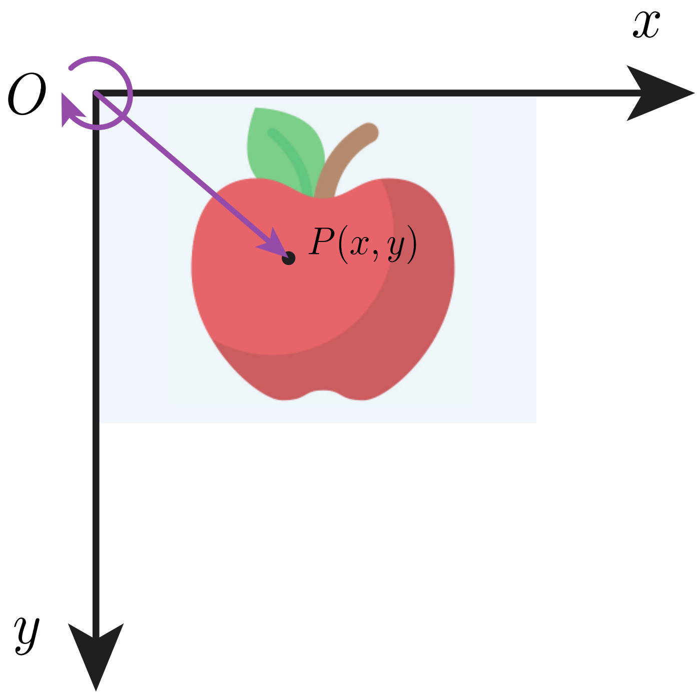
Continuous and
right-handed



Coordinates:
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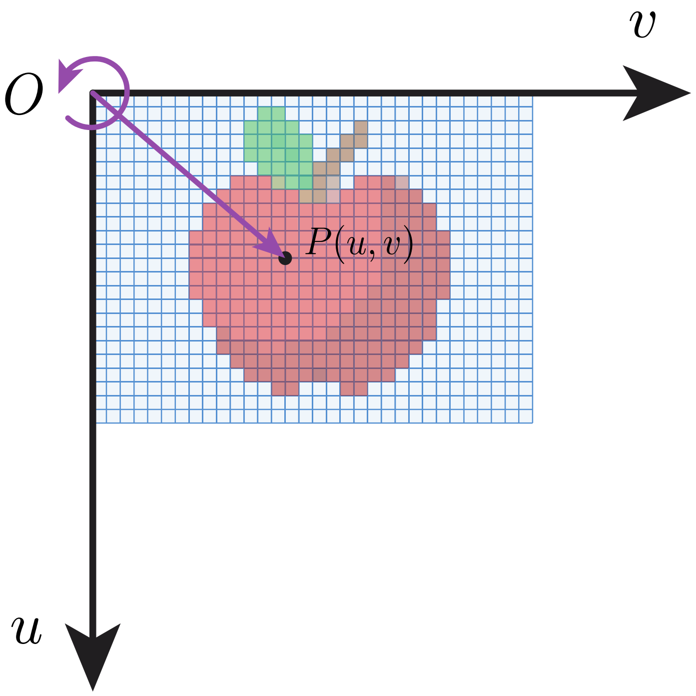
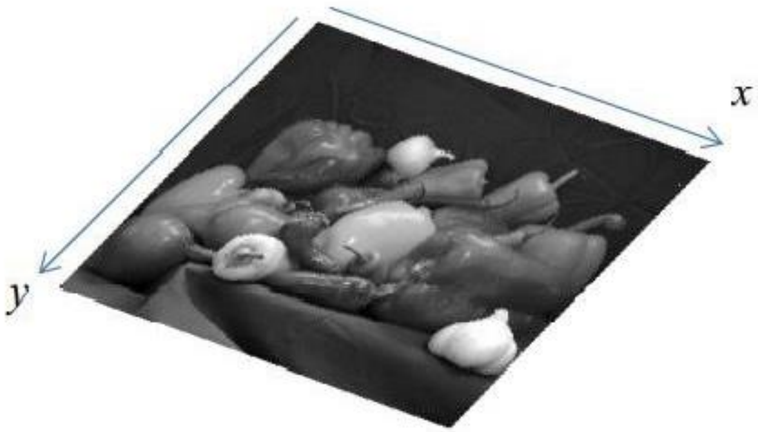


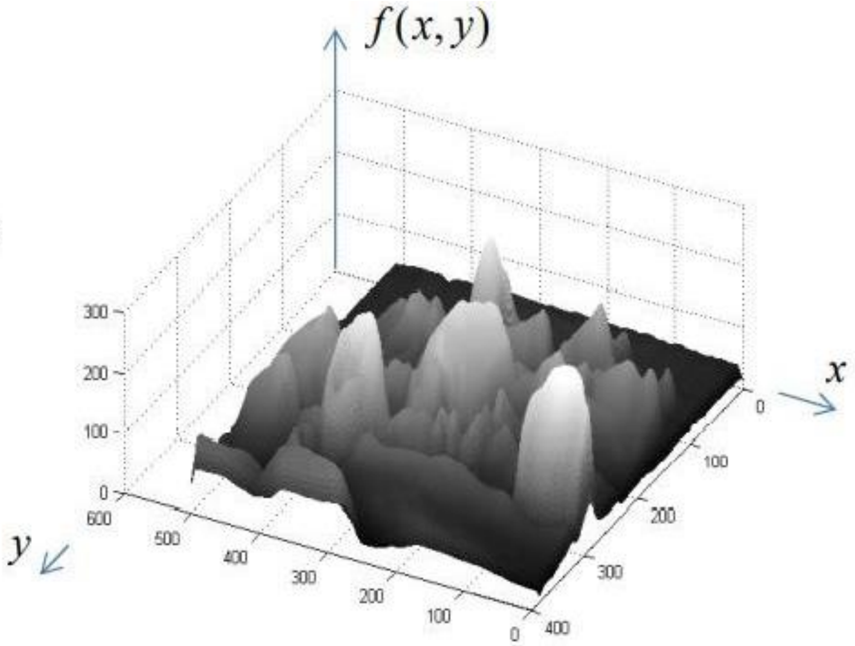
Coordinates:
Discrete and
right-handed

Image as a function

- A painted grayscale image can be considered as a continuous function over a region in $\Omega \in \mathbb{R}^2$, then at each point (x, y) point we get a gray value g which we can describe by a number $g = f(x, y) \in \mathbb{R}$:

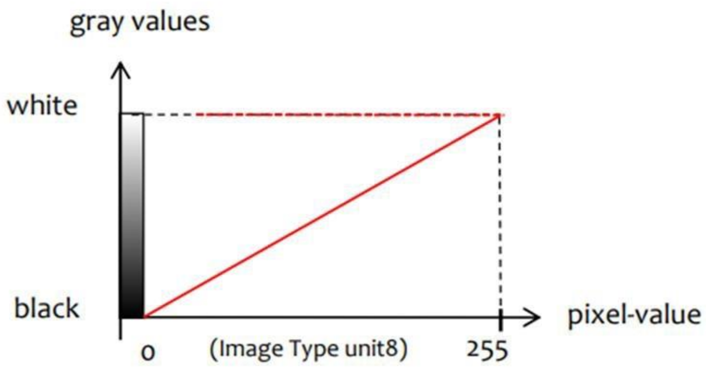
$$f : \Omega \in \mathbb{R}^2 \to \mathbb{R}, \quad (x, y) \to f(x, y)$$





Pixel values / Grayscale values

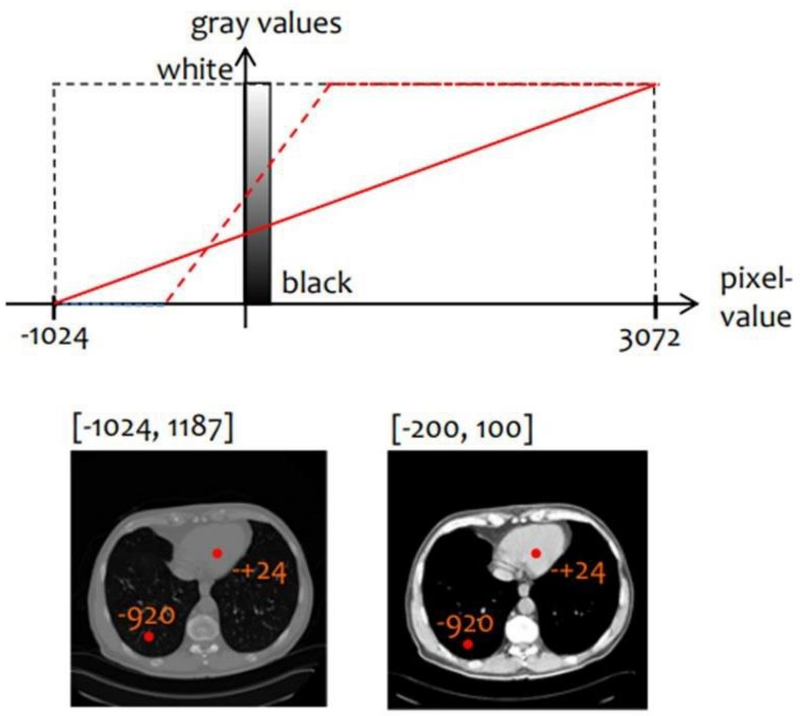
- Many digital images use **256 possible gray levels**, ranging from black to white. This corresponds to the number of values representable with **8 bits (1 byte)**.
- Convention:
  - 0000 0000 (decimal 0) → black
  - 1111 1111 (decimal 255) → white
  - Intermediate values encode grayscale levels in between.



Pixel values / Grayscale values

- In many applications, pixel values represent physical quantities, not just brightness (unlike in standard grayscale photography). For example:
  - Radiation absorption (e.g., X-ray / CT)
  - Magnetization (e.g., MRI)
  - Other modality-specific signals

- Implications:
  - Pixel values may lie in arbitrary value ranges (and not 0-255)
  - For storage and visualization, we still prefer a standard value range [0, 255]
  - Therefore, pixel values must be mapped (normalized) to this range



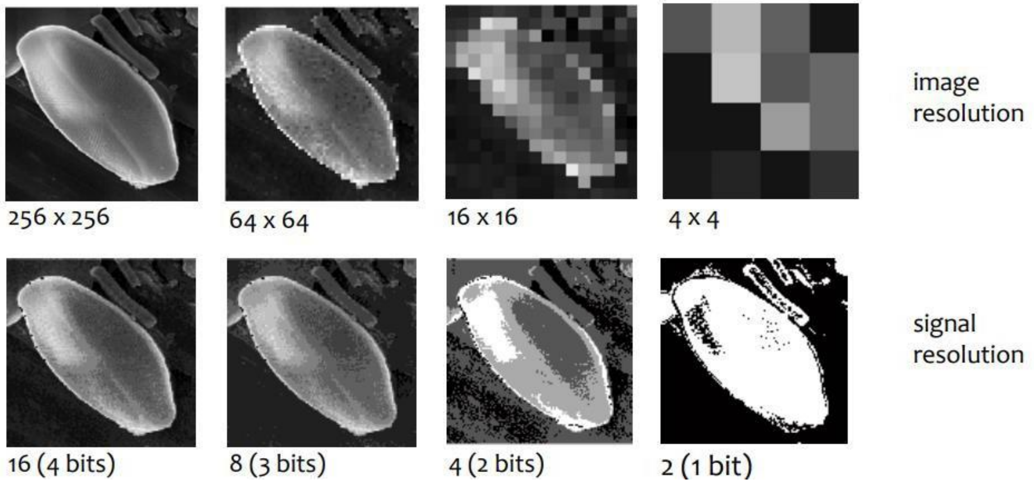
In CT data, pixel values typically range from –1024 to 3072. If only a selected portion of this range is mapped to the display grayscale range, this process is called windowing.

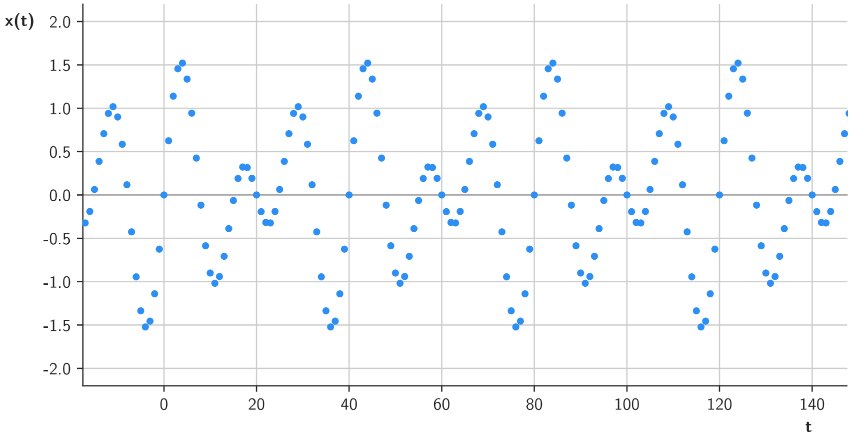
How many grayscale levels?

- **Image depth** (bit depth):
  - Number of bits used to represent the intensity or color of each pixel
  - Determines how many different shades (or colors) can be displayed
- **Integer** vs. **floating-point** representation
  - Image data is often stored as integers (e.g., 8-bit, 16-bit)
  - For image processing, floating-point representations (e.g., 16, 32, 64 bit) are common
  - Typical normalization for floating-point images: $p \in [0,1]$, with 0 black, 1 = white
- Other bit depths and representations
  - 16-bit images: Used in scientific and medical imaging for higher dynamic range
  - Lower bit depths (e.g., <8-bit): Used to reduce memory requirements (e.g., in JPEG format)
  - Floating-point images: Used during processing (e.g., filtering, averaging, normalization)
- Note: Bit depth directly affects the memory required to store an image in memory

Digitalization / discretization

- Digitalization involves two steps:
  - Spatial quantization (sampling)
  - Intensity quantization.
- Quantization means that values are restricted to a finite set of permissible levels (e.g., integers instead of continuous values)





Why 8-bit images are common

- The human visual system can distinguish only a limited number of gray levels (≈ 80 shades, depending on conditions)
- 8-bit (256 levels) is therefore sufficient for most visual applications

Task: Try to connect this to time-series signals.

Intensity quantization

- In grayscale images, each sampled intensity value must be assigned to the **nearest available gray level**
- This introduces an approximation error, known as **quantization error**
- The error decreases as the number of gray levels increases
- However, more gray levels require **higher bit depth** and therefore more **storage**

64 shades of gray

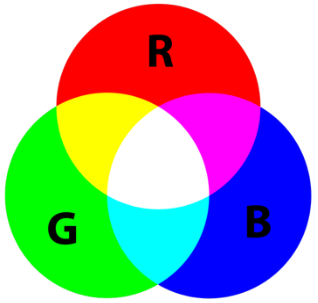
8 shades of gray

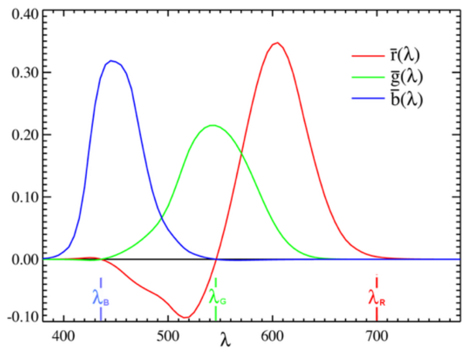
Color representation

Colors and color spaces

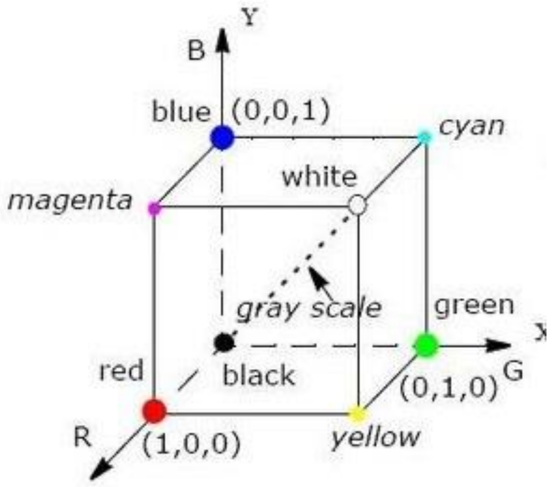
The RGB color space

- Trichromatic theory of color vision:
  - Human color perception can be approximated by combining three primary colors: red, green, and blue
  - By varying their intensities, a wide range of colors can be represented → RGB color space
- CIE 1931 RGB color space
  - Device-independent standard defined by the International Commission on Illumination (CIE) in 1931
  - Based on three primary colors (R, G, B) at wavelengths 700 nm, 546 nm, 436 nm
- Notes:
  - Many other, device-specific RGB color spaces exist!
  - The diagonal of the RGB cube represents the gray scale axis (R=G=B)





These curves do not match the spectral sensitivity curves of humans for various reasons (e.g., color light generation in 1931)

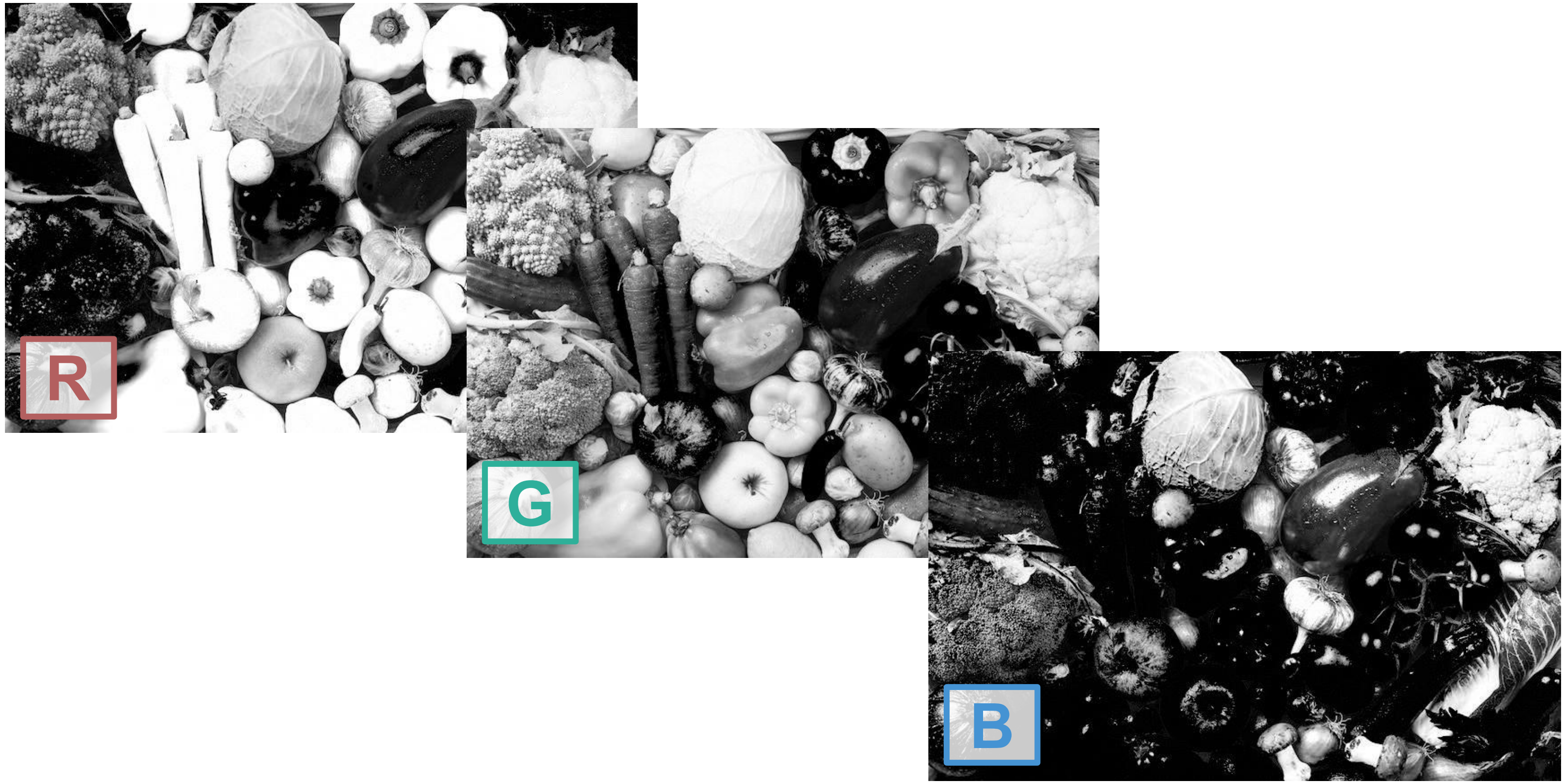


Example: RGB color space

- In an RGB color image, each pixel contains three values (R, G, B); therefore, a color image can be represented as three grayscale images.

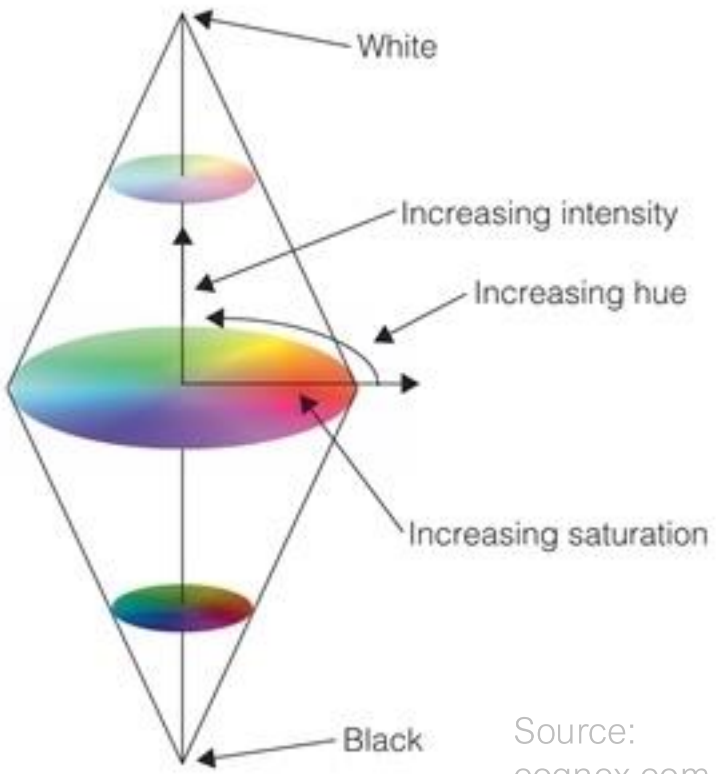
Task: Can you guess which of the below images represent R, G or B?

$$p = \begin{pmatrix} 255 \\ 251 \\ 0 \end{pmatrix}$$



The HSI color space (Hue, Saturation, Intensity)

- Limitations of RGB:
  - RGB is well suited for image acquisition and display (cameras, monitors)
  - However, interpreting colors in RGB is not intuitive.
- Motivation:
  - Represent color in a way that aligns with human perception
  - Separates color information (H, S) from brightness (I)
- Components of HSI:
  - H: Hue (type of color, e.g., red, yellow, magenta)
  - S: Saturation (purity or vividness of the color)
  - I: Intensity (brightness of the color)



Source: cognex.com

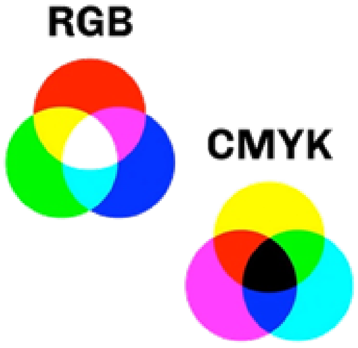
The HSI color space (Hue, Saturation, Intensity)

- In the HSI space, the color can be manipulated without affecting the brightness or saturation of color.

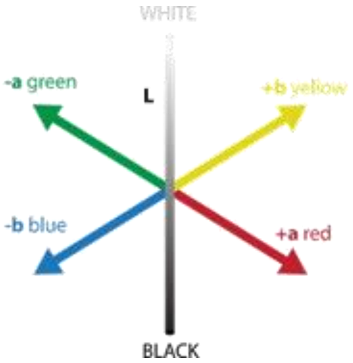
Original Hue-Shift 1/6 Hue-Shift 2/6 Hue-Shift 5/6

Other color spaces

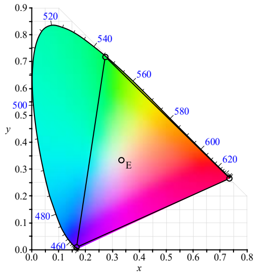
- CMYK (Cyan, Magenta, Yellow, Key / black):
  - Represents colors by subtracting cyan, magenta, and yellow from white light.
  - Hence, it is a subtractive color model (contrary to RGB, or HSI)
  - Includes a black channel (K, key) to improve the reproduction of dark colors
  - Standard color model used in color printing and graphic design.



- L*a*b* (CIELAB):
  - A device-independent color space designed to approximate human vision more accurately than RGB or CMYK.
  - Consists of three components: L* (lightness), a* (green to red), and b* (blue to yellow).
  - Widely used in color science and for color difference calculations



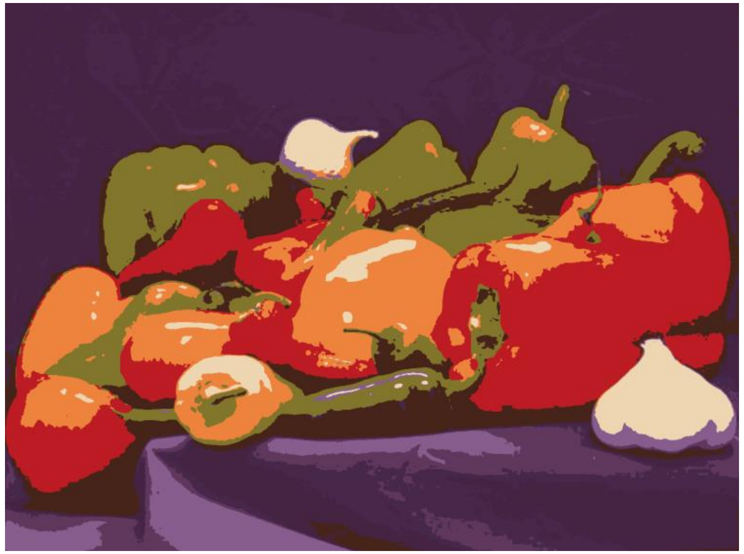
- XYZ (CIE 1931 XYZ):
  - A standard based on human color perception
  - Represents colors using tristimulus values X, Y, and Z, which correspond to the stimulation of the three types of cones in the human eye.



Reduction of color spaces: Dithering

- Like the gray scale, also the color scale can be reduced
- Simple nearest-neighbor color-mapping yields color bands
- Dither: Method to avoid such pattern by adding noise

Input: $2^{8 \cdot 3} = 2^{24}$ = 16.7 Mio colors
Goal: 10 target colors



Method 1: Nearest neighbor mapping results in color bands and a comic-like appearance.

Demo: Photoshop

Method 2/3: Introduce random noise to avoid bands (second method: use diffusion approach to for better results)

Reduction of color spaces: Color to grayscale conversion

- Goal: convert the three RGB values (per pixel) to one grayscale intensity value.
- There are different models for this conversion

Input image

value

$$V = \max(R,G,B)$$

luminance

$$L = \frac{\max(R,G,B)+\min(R,G,B)}{2}$$

intensity

$$I = \frac{R+G+B}{3}$$

weighted intensity

$$I = 0.3\cdot R + 0.6\cdot G + 0.1\cdot B$$

Alternative ways to represent digital images

Alternative image representations: Pyramidal images

- Alternative data structure for images
- Hierarchical structure of images created by successive subsampling operations (and prior filtering)
- Idea: Maintain different resolutions of the same image
- Advantages:
  - Memory optimization: Only load parts into memory that are currently required
  - Enables multi-resolution analysis
- Disadvantage:
  - Requires more memory

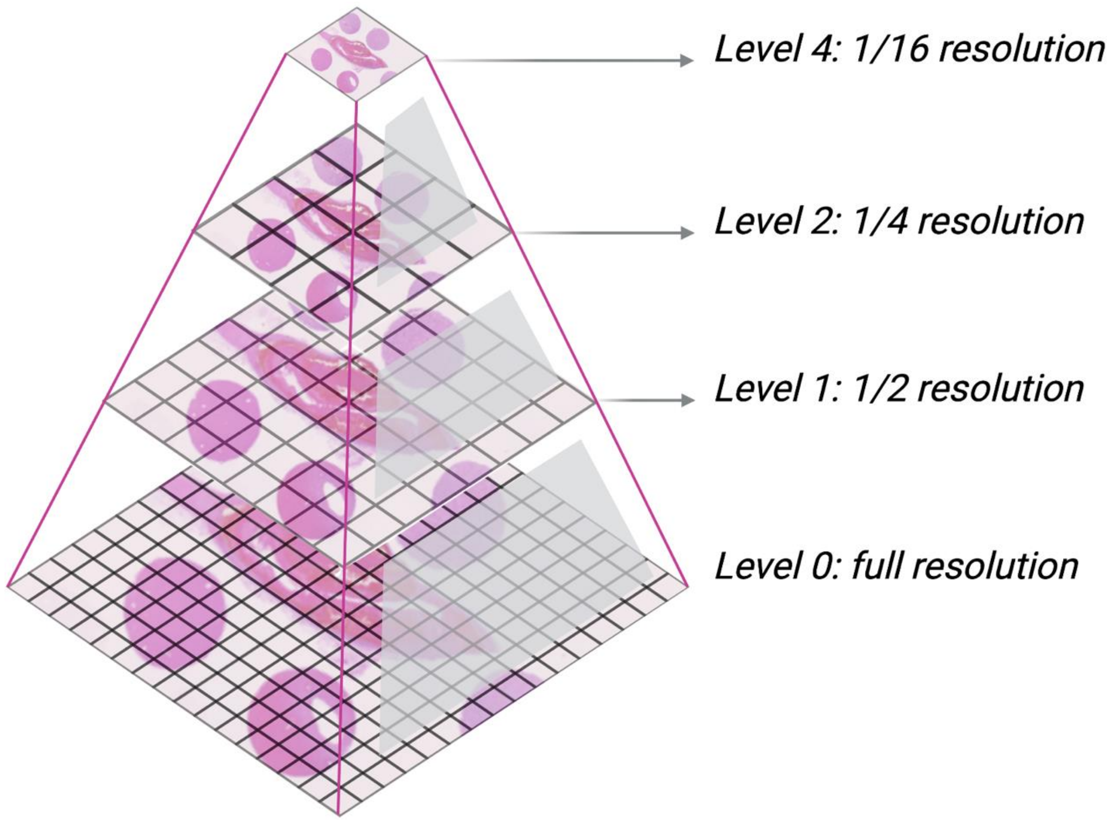
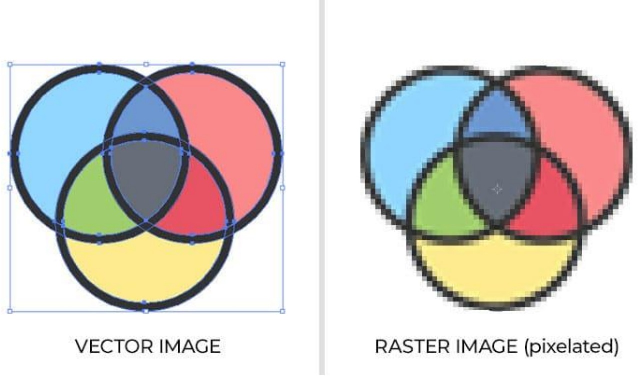


Illustration: Leoni Grossmann

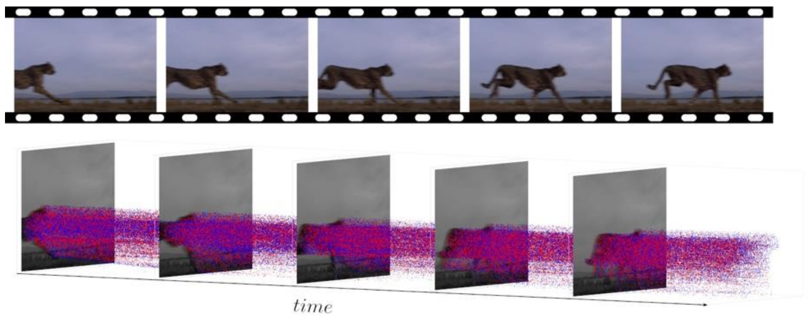
Alternative image representations

- Vector graphics (instead of rasterized data)
  - Instead of representing an image as a grid of pixels, describe it using geometric primitives such as points, lines, curves, or polygons.
  - More on this later…



Source: impact-nw.com

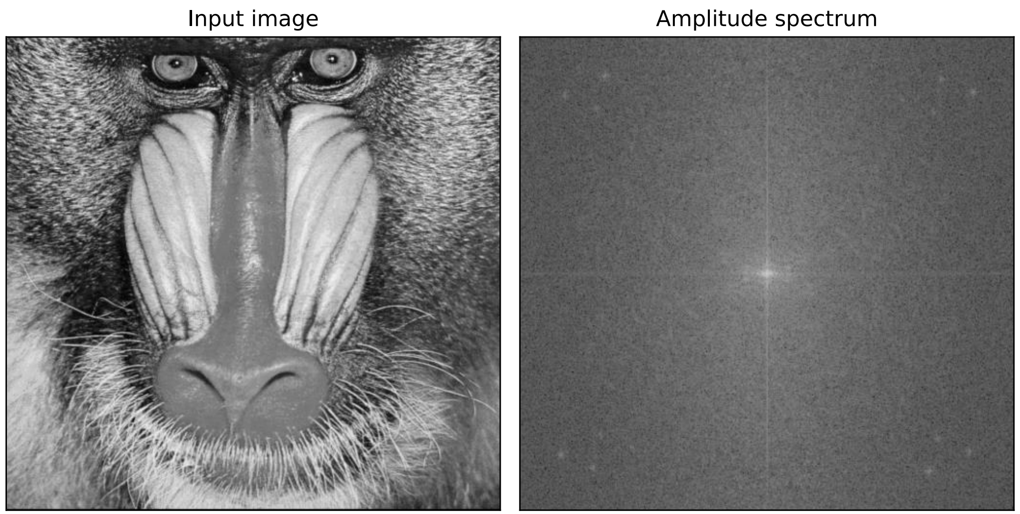
- Sparse / event-based representations
  - Not all images require a dense pixel grid.
  - Sparse images store only relevant or non-zero pixels.
  - Event-based cameras capture changes over time → Data is stored as a list of events, not a full matrix.
  - Explanatory video, by LUCID Vision Labs



Event-based vs. dense imaging: only temporal changes are recorded, enabling higher effective sampling rates. Source: Wikimedia

Alternative image representations

- Frequency-domain representations
  - Represent images using Fourier coefficients.
  - You guessed it: More on this later… ☺



→ Image representations depend on the application, requirements, and available resources / hardware.

→ In this module, we focus on digital images in a dense pixel representation, as introduced earlier.

The digital image processing system

The digital image processing system

A complete digital image processing system consists of hardware (equipment) and software (algorithms and programs) that together perform the following tasks:

1. **Image acquisition**: Capture an image and, if necessary, convert it to digital form using an analog-to-digital converter (ADC)
2. **Storage**: Store the image temporarily (in memory) or permanently (on disks or SSDs)
3. **Processing**: Manipulate and analyze the image (e.g., enhancement, filtering, segmentation)
4. **Display**: Visualize the image, typically via a display system (historically involving DACs)